

Computational Fluid Dynamics of a Buoyancy-Driven Chimney System

Overview:

The project modelled buoyancy-driven flow through a two-dimensional heat chamber-chimney system. Hot air rises through the channel and chimney while cooler air is drawn in through lateral inlets, a natural-convection configuration modelled with the Boussinesq approximation. This heat-chamber-chimney mechanism is shown below:

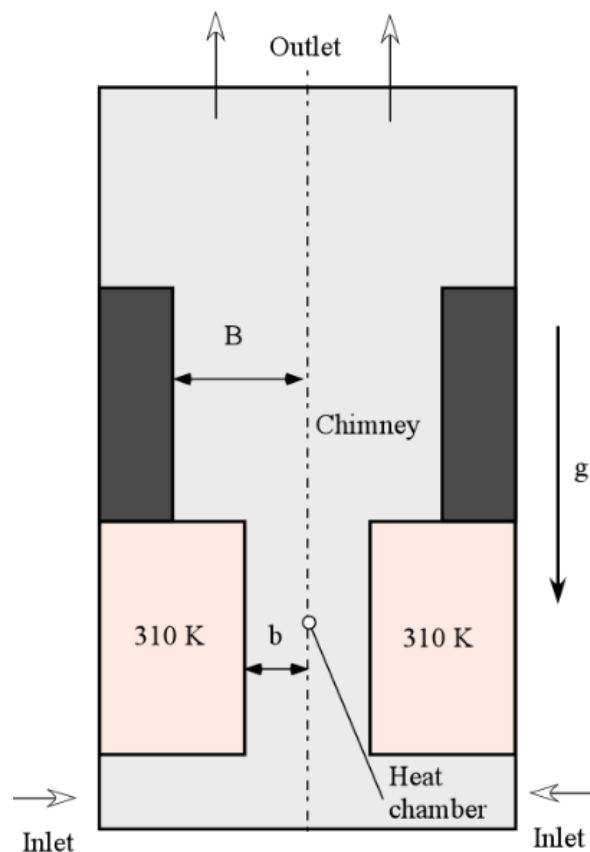


Figure 1: Heat Chamber-Chimney Mechanism

The aim was to build and validate the CFD model, establish a grid-independent mesh and critically compare the pressure-based and density-based solves together using the turbulence-modelling options for the flow.

A two-dimensional geometry model of the heat chamber-chimney was created in ANSYS Design Modeler and simulated in ANSYS Fluent using a steady state analysis followed by a transient analysis. A total of nine linked studies were conducted. The Reynolds, Rayleigh and Mach numbers were hand calculated throughout to verify the modelling assumptions.

Key Results:

The face meshing feature was used to categorise the structure with quadrilateral elements, giving a orthogonal quality of 1.0 and negligible skewness, confirming numerical stability. The chosen Mesh from the grid independence study is shown below:

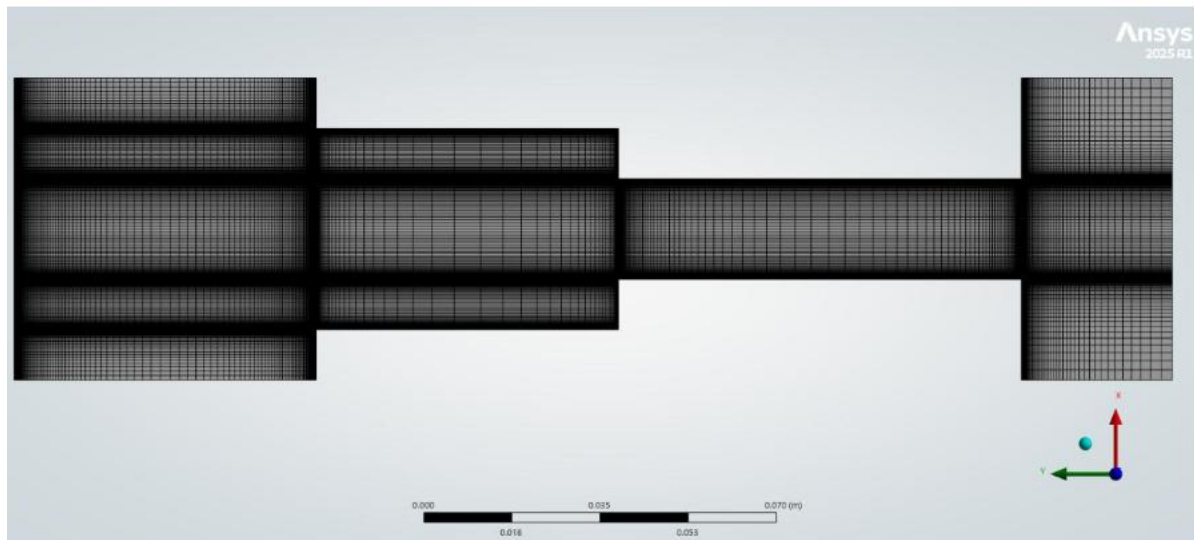


Figure 2: Longitudinal View of chosen Mesh from Grid Independence Study

The model captured a central buoyancy-driven high velocity jet rising through the channel and chimney, with a peak velocity of 0.496m/s. This is shown in **Figure 3**.

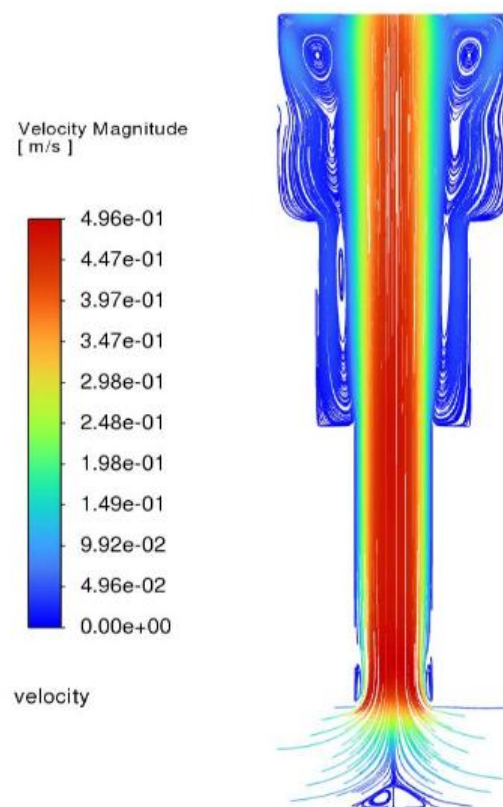


Figure 3: Velocity Magnitude Pathlines showcasing the

The solver assumptions were verified against computed Rayleigh, Reynolds and Mach numbers.

The pressure-based and density-based solvers produced near identical velocity magnitudes, however, the density-based solver required ~88% more computation time. The pressure-based solver converged smoother and produced a more developed velocity pathline, as illustrated in **Figure 4** & **Figure 5**.

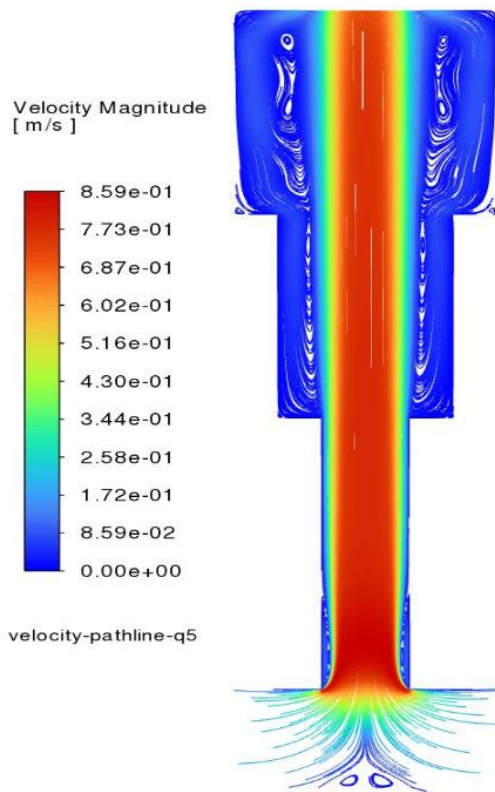


Figure 4: Velocity Pathline for Pressure-Based Solvers

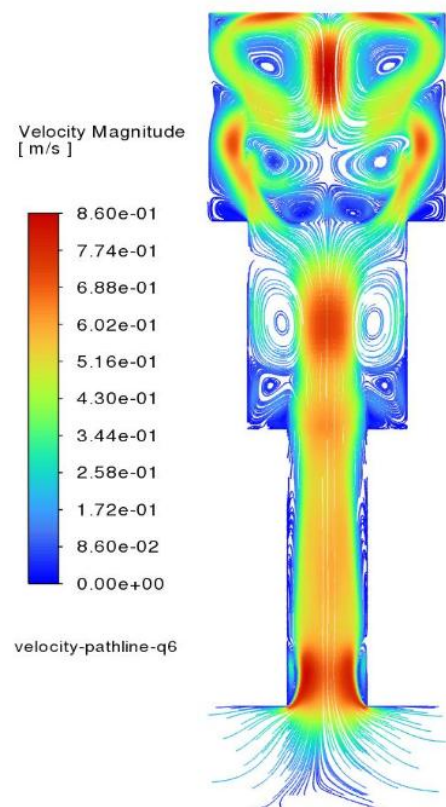


Figure 5: Velocity Pathline for Density-Based Solvers

Note: The boundary conditions were changed when conducting the Pressure v Density based solver study, thus explaining the difference between the velocity magnitudes found in **Figure 3** and **Figure(s) 4,5**.

The flow operates within a highly turbulent regime, characterised by a pulsating central jet and the shedding of multiple eddies. A DNS was shown to be impractical, so a Large Eddy Simulation with the Smagorinsky-Lilly sub-grid model was recommended over RANS models.